

DANIEL SUN

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EDUCATION

University of Toronto — M.Sc. in Statistics Sept 2026 - Present
Supervisor: Prof. Lisa Strug

University of Toronto — Hon. B.Sc. in Statistics and Economics Sept 2022 – Apr 2026
Data Sciences Institute Scholar Award (\$7200) *Major GPA: 3.95*

Statistical Machine Learning, Time Series Analysis, Statistical Computation, Methods of Multivariate Statistics, Surveys, Sampling and Observational Data, Applied Econometrics, Real Analysis, Game Theory

RESEARCH EXPERIENCE

Machine Learning Methods for Binding and Potency Prediction Aug 2026 - Present
Supervisors: Prof. Lisa Strug *SickKids*

- Implementing and benchmarking ML tools for efficacy, potency & binding affinity predictions.
- Literature review of existing applied statistics and ML tools.

GraphCoVaR: Systemic Fire-Sale Risk in Financial Networks Jun 2026 - Aug 2026
Supervisors: Prof. Ing-Haw Cheng and Prof. Goutham Gopalakrishna *Rotman FinHub*

- Scaled the Greenwood-Landier-Thesmar fire-sale model to 136k investors using sparse computation to avoid a 18.5-billion entry overlap matrix.
- Built and validated null models with Monte Carlo, bootstrap, and sensitivity analyses to measure excess crowding.
- Led reproducibility testing of a 31,000-line package, identifying sampler bias and deployment-blocking defects.

Visitor Modelling at the Art Gallery of Ontario May 2026 - Aug 2026
Supervisor: Prof. Jessica Gronsbell *Data Sciences Institute*

- Performed EDA on categorical survey data and used minilm-l6-v2 to group free-response text
- Created visualizations and performed K-modes to compare with the Gallery's internal personas
- Developed and deployed an interactive [Shiny webapp](#) survey results to non-technical stakeholders.

Interactive Coding Modules for Biological Data Analysis Feb 2026 - Present
Supervisors: Prof. Alexandra Malinowski and Prof. Gracia Dong *Dept. Stats, UofT*

- Designed and developed interactive [Python](#) and [R modules](#) for Principal Component Analysis applied to biological data.
- Programmed computational tools for sample size calculations and systematic review visualizations, including forest plots and odds ratio evaluation.

Generative AI Usage Among Undergraduate Statistics Students Jul 2025 - Apr 2026
Supervisors: Prof. Samantha-Jo Caetano and Prof. Junhao Zhu *Dept. Stats, UofT*

- Surveyed undergraduate statistics students on generative AI usage; currently preparing a manuscript of the findings for academic publication.
- Conducting a literature review of generative AI tools and supports for undergraduate statistics education, synthesizing interdisciplinary findings.

Statistics Course Policy Research Nov 2025 - Apr 2026
Supervisor: Prof. Samantha-Jo Caetano *Dept. Stats, UofT*

- Performed data cleaning and descriptive statistical analysis in R to support course policy research; produced an academic poster and conference presentation.

R Modules for Regression Instruction

Dec 2024 - Feb 2025

Supervisor: Prof. Katherine Dagnault

Dept. Stats, UofT

- Developed self-paced R modules using Broom and learnR to translate regression concepts into interactive instructional content.
- Conducted quality assurance testing of peer-created modules, providing detailed written feedback for optimization.

PRESENTATIONS/TALKS

Visitor Behaviour Modelling at the Art Gallery of Ontario

August 2026

Data Sciences Institute Conference, University of Toronto — Best Oral Presentation Award

Clustering, Survey Methodology

Course Policy Research Findings

December 2025

MacPherson Institute Innovations in Education Conference, McMaster University

Applied Data Analysis

TEACHING

Surveys, Sampling and Observational Data (STA304)

Fall 2026

Teaching Assistant at the University of Toronto

Probability, Statistics and Data Analysis II (STA238)

Winter 2027

Teaching Assistant at the University of Toronto

WORK EXPERIENCE

City of Toronto / University of Toronto

Aug 2025 - Sept 2025

Statistical Consultant

- Analyzed public Wi-Fi usage data across Toronto by developing and reviewing Python scripts (Pandas, GeoPandas, scikit-learn) contributing to a peer-reviewed academic study.
- Validated theoretical methodology and recommended unsupervised methods to generate actionable geospatial insights for municipal decision-makers.

Atlantis AI

May 2025 - Aug 2025

AI Agent Engineer, Internship

- Designed a cost-minimizing linear programming algorithm for a payroll management system and implemented it in Python with HiGHS.
- Developed an AI agent using LangChain to wrap the optimizer, and analyzed completion and accuracy metrics via LangSmith.
- Set up, deployed, and automated updates for a live web application using Netlify.

zofiQ Inc.

Feb 2025 - May 2025

Machine Learning Assistant

- Contributed to a self-monitoring ensemble AI agent that improved classification accuracy of IT tickets.
- Designed concept drift detection to monitor data distribution changes and trigger model retraining.

Platanus Group

May 2024 - Aug 2024

Data Analyst, Internship

- Extracted and cleaned historical S&P 500 data from PostgreSQL databases and Excel for use in modelling.
- Engineered RSI and volatility features and applied XGBoost to generate trading signals in R.

- Integrated model outputs and SHAP values back into PostgreSQL to support strategy evaluation and dashboards.

SKILLS/HOBBIES

Programming Languages

Python, R, SQL, STATA, Git, $\text{\LaTeX}$

Spoken Languages

English, French, Mandarin

Hobbies

Hiking, Comedy, Cooking